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"""
Number Guessing Game & Word Counter
A beginner-friendly CLI program with:
  - A number guessing game using
random numbers
  - A file-based word counter tool
  - Input validation, functions,
loops, and file handling
"""

import random
import string
from collections import Counter


# ---------- Helpers ----------
def get_int(prompt, min_value=None,
max_value=None):
    """Keep asking until the user
enters a valid whole number in
range."""
    while True:
        value = input(prompt).strip()
        try:
            number = int(value)
        except ValueError:
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            print("Invalid input.
Please enter a whole number.")
            continue

        if min_value is not None and
number < min_value:
            print(f"Number must be at
least {min_value}.")
        elif max_value is not None
and number > max_value:
            print(f"Number must be at
most {max_value}.")
        else:
            return number


def get_choice(prompt,
valid_choices):
    """Keep asking until the user
enters one of the valid choices."""
    while True:
        choice =
input(prompt).strip().lower()
        if choice in valid_choices:
            return choice
        print(f"Invalid choice.
Please pick one of: {',
'.join(valid_choices)}")
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# ---------- Number Guessing Game ----------
def play_game():
    print("\n--- Number Guessing Game ---")
    print("Choose a difficulty:")
    print("1. Easy   (1-50,  10 attempts)")
    print("2. Medium (1-100, 7 attempts)")
    print("3. Hard   (1-200, 5 attempts)")
    level = get_choice("Choose (1-3): ", ["1", "2", "3"])

    if level == "1":
        upper, max_attempts = 50, 10
    elif level == "2":
        upper, max_attempts = 100, 7
    else:
        upper, max_attempts = 200, 5

    secret = random.randint(1, upper)
    attempts = 0
    print(f"\nI'm thinking of a number between 1 and {upper}.")
    print(f"You have {max_attempts} attempts. Good luck!")
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    while attempts < max_attempts:
        guess = get_int(f"Attempt
{attempts + 1}/{max_attempts} - Your
guess: ", 1, upper)
        attempts += 1

        if guess == secret:
            print(f"Correct! You
guessed it in {attempts}
attempt(s).")
            return attempts
        elif guess < secret:
            print("Too low!")
        else:
            print("Too high!")

    print(f"Out of attempts. The
number was {secret}.")
    return None


def guessing_game():
    best_score = None
    while True:
        result = play_game()
        if result is not None and
(best_score is None or result <
best_score):
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            best_score = result
        if best_score is not None:
            print(f"Your best score
this session: {best_score}
attempt(s)")

        again = get_choice("Play
again? (yes/no): ", ["yes", "no",
"y", "n"])
        if again in ("no", "n"):
            break


# ---------- Word Counter ----------
def read_file():
    """Ask for a file name until a
readable file is found (or user
cancels)."""
    while True:
        filename = input("Enter the
text file name (or 'back' to return):
").strip()
        if filename.lower() ==
"back":
            return None
        try:
            with open(filename, "r",
encoding="utf-8") as file:
                return file.read()
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        except FileNotFoundError:
            print(f"File '{filename}'
was not found. Try again.")
        except PermissionError:
            print("You don't have
permission to read that file.")
        except UnicodeDecodeError:
            print("That file isn't a
plain text file. Try another one.")


def create_sample_file():
    """Create a small sample file so
the word counter can be tested."""
    sample = (
        "Python is fun. Python is
powerful.\n"
        "Learning Python step by step
makes programming easy.\n"
    )
    with open("sample.txt", "w",
encoding="utf-8") as file:
        file.write(sample)
    print("Created 'sample.txt' in
the current folder.")


def word_counter():
    print("\n--- Word Counter ---")
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    print("1. Count words in a file")
    print("2. Create a sample.txt
file to test with")
    print("3. Back")
    choice = get_choice("Choose (1-
3): ", ["1", "2", "3"])

    if choice == "2":
        create_sample_file()
        return
    if choice == "3":
        return

    text = read_file()
    if text is None:
        return

    lines = text.splitlines()
    # Remove punctuation and make
lowercase so "Python," and "python"
match
    cleaned =
text.translate(str.maketrans("", "",
string.punctuation)).lower()
    words = cleaned.split()

    print("\n--- Results ---")
    print(f"Lines      :
{len(lines)}")
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    print(f"Words      :
{len(words)}")
    print(f"Characters :
{len(text)}")
    print(f"Unique words:
{len(set(words))}")

    if words:
        print("\nTop 5 most common
words:")
        for word, count in
Counter(words).most_common(5):
            print(f"  {word}:
{count}")
    else:
        print("The file is empty.")


# ---------- Main menu ----------
def main():
    print("=== Number Guessing Game &
Word Counter ===")
    while True:
        print("\nMain Menu")
        print("1. Number Guessing
Game")
        print("2. Word Counter")
        print("3. Exit")
        choice = get_choice("Choose
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an option (1-3): ", ["1", "2", "3"])

        if choice == "1":
            guessing_game()
        elif choice == "2":
            word_counter()
        else:
            print("Thanks for
playing. Goodbye!")
            break


if __name__ == "__main__":
    main()
```